Gas Leaks Hint Mars May Still Be Volcanically Alive

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Mars may not be as dead as it looks. Strange puffs of gas rising from cracks near a giant Martian volcano suggest heat and volcanic activity could still be hiding deep under the surface. Scientists found unusual amounts of hydrogen chloride gas above cracks on Alba Mons, a huge volcano. The gas kept showing up in early summer, when it normally almost disappears, always over the same cracks. An international team led by Alexander Trokhimovskiy thinks the gas could be coming from hot **magma** underground. The gas was first spotted by the European Space Agency's ExoMars Trace Gas **Orbiter** in 2021. Close-up pictures showed narrow cracks that look quite fresh, some stretching hundreds of metres. Scientists think bitterly cold Martian winters, when temperatures can fall to minus 120°C, let ice seal the cracks shut. As temperatures rise, the ice turns straight into **vapour**, letting underground gases escape. Mars is home to the largest volcanoes in the Solar System. But nobody has ever seen a new **eruption** on Mars, so whether it is still **volcanically** active remains an open question.

Word Treasure

magma -- Hot melted rock deep inside a planet.
vapour -- A gas made when a liquid or ice turns into tiny drops in the air.
Orbiter -- A spacecraft that circles around a planet to study it.
eruption -- When a volcano bursts open and lets out hot rock and gas.
volcanically -- To do with volcanoes.

Background you need

A volcano is a mountain that can let out hot melted rock, called magma, and gases from deep inside a planet.

Mars looks cold and quiet today, but it has the biggest volcanoes in the Solar System.

Scientists use spacecraft called orbiters to take pictures and measurements from above Mars.

Break it down

WHO: An international team of scientists led by Alexander Trokhimovskiy.
WHAT: Strange puffs of gas over cracks on Mars may mean the planet is still volcanically alive.
WHERE: Above cracks on Alba Mons, a huge volcano on Mars.
WHY: The gas may be coming from hot magma deep underground.

Why it matters

If you ever visit Mars in a story or a game, this helps you picture a planet that may still be warm and active under its cold surface.

Test what you remember

1. What gas did scientists find over the cracks?

- (A) Carbon dioxide
- (B) Nitrogen
- (C) Oxygen
- (D) Hydrogen chloride

2. Over which volcano were the cracks found?

- (A) Mount Etna
- (B) Mount Everest
- (C) Alba Mons
- (D) Olympus Mons

3. When did the gas keep showing up?

- (A) In deep winter
- (B) Only in storms
- (C) Only at night
- (D) In early summer

4. What spacecraft first spotted the gas?

- (A) The Hubble Telescope
- (B) The James Webb Telescope
- (C) The ExoMars Trace Gas Orbiter
- (D) A Mars rover

5. How cold can Martian winters get?

- (A) Minus 20°C
- (B) Minus 120°C
- (C) Minus 200°C
- (D) Minus 60°C

6. Has anyone ever seen a new eruption on Mars?

- (A) Only on TV
- (B) No, never
- (C) Yes, once
- (D) Yes, many times

Answer key (try the quiz first!): 1: D 2: C 3: D 4: C 5: B 6: B

Questions to think about

1. The research team led by Alexander Trokhimovskiy: It thinks the gas could be coming from hot magma underground and that Mars may still be volcanically active.

MY THOUGHTS (at least 20 words):